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# Screening of natural compounds for identification of novel inhibitors against β-lactamase CTX-M-152 reported among *Kluyvera georgiana* isolates: An *in vitro* and *in silico* study

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ABSTRACT

Multidrug resistance due to the expression of extended spectrum β-lactamases (ESBLs) by bacterial pathogens is an alarming health concern with huge socio-economic burden. Here, 102 bacterial isolates from Wastewater treatment plants (WTPs) were screened for resistance to different antibiotics. Kirby-Bauer method and phenotypic disc confirmatory test confirmed the prevalence of 20 ESBLs. Polymerase chain reaction-based detection confirmed 11 *bla*$_{CTX-M}$ positive bacterial isolates. Genotyping of bacterial isolates by 16S rRNA gene sequencing showed the dissemination of *bla*$_{CTX-M}$ in *Escherichia fergusonii*, *Escherichia coli*, *Shigella* sp., *Kluyvera georgiana* and *Enterobacter* sp. Amongst *Kluyvera georgiana* isolates, two were harboring *bla*$_{CTX-M-152}$. The 3D model of CTX-M-152 protein was generated using SwissProt and characterized by Ramachandran plot and SAVES. A library of natural compounds was screened to identify novel CTX-M-152 inhibitor(s). High-throughput virtual screening (HTVS), standard precision (SP) and extra precision (XP) docking led to the identification of five natural compounds (Naringin dihydrochalcone, Salvianolic acid B, Inositol, Guanosine and Ellagic acid) capable of binding to active site of CTX-M-152. Futher, characterization by MM-GBSA (Molecular Mechanism General Born Surface Area), and ADMET (Adsorption, Distribution, Metabolism, Excretion and Toxicity) showed that Ellagic acid was the most potent inhibitor of CTX-M-152. Molecular dynamics simulation also confirmed that Ellagic acid form a stable complex with CTX-M-152. The ability of Ellagic acid to inhibit growth of bacteria harboring CTX-M-152 was confirmed by MIC (Minimum Inhibitory Concentration; broth dilution method) and Zone of Inhibition (ZOI) studies with respect to Cefotaxime. The identification of a novel inhibitor of CTX-M-152 from a natural source holds promise for employment in the control of bacterial infections.

## 1. Introduction

Antibiotics play a vital role in fighting infections caused by bacteria and other microbes. Owing to rise in the usage of antibiotics, some of disease-causing microbes have become remarkably resilient against antibiotics which has raised concern regarding global health [1]. With rise in resistance, it becomes a challenging task to control the infections caused by resistant bacteria that resulted in long-lasting sickness and larger menace of fatality [2]. The clinical and environmental discharges are pivotal in enhancing the Multidrug resistance (MDR) phenomenon

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[3,4]. Facilitating horizontal gene transfer [5], consumption of water by human beings from natural water reservoirs that are often contaminated results in the spread of resistance [6]. Though, the biological treatment of wastewater trend to decrease the quantity of bacterial overload, these processes still lacks in eliminating the antibiotic and other resistant determinants (heavy metal and biocide), and hence the remnants enter in different ecosystems through food cycles, wastewater discharge and bio-sludge [7]. The wastewater treatment plants (WTPs) are a hub that hosts resistant bacteria harboring predominantly disseminated Extended spectrum β-lactamases (ESBL's) such as *bla*$_{TEM}$, *bla*$_{SHV}$ and recently emerged *bla*$_{CTX\text{-}Ms}$. Other genes imparting resistance to different classes of antibiotics such fluoroquinolone resistant genes (*qnrA*, *qnrB* and *qnrS*), macrolide resistant genes (*erm*(B)), sulfonamides resistant genes (*sul*(I) and *sul*(II)) and tetracycline resistant genes (*tet*(O) and *tet* (W)) have also been reported in WTPs [8–10].

β-lactamases are categorized in four classes (class A-D) according to Ambler's classification [11]. Class A β-lactamases (or 2be according to Bush-Jacoby-Medeiros) degrade penicillin by cleaving the amide bond present in the structure of β-lactam antibiotics [12]. CTX-Ms are the most widely disseminated members of ESBL's family. Although, CTX-M-1 was the first variant of CTX-M family reported in 1990's, more than 172 variants of CTX-M have been reported so far from different parts of the world. These ESBLs are a major threat to the available medical regime as they confer resistance to presently used Amino-penicillins (Ampicillin, Amoxicillin), Carboxypenicillins (Carbenicillin, Tricarcillin), Ureidopenicillin (Piperacillin), Cephalosporins (Cephalothin, Cephaloridine, Cefuroxime), oxi-imino Cephalosporins (Cefotaxime, Ceftriaxone), Cefepime, Cefpirome, etc [13]. A novel ESBL's variant CTX-M-152 gene has been recently reported in *Kluyvera* sp. [13]. As *bla*$_{CTX\text{-}M\text{-}152}$ is a member of CTX-M-25 group (a less disseminated variant of CTX family), it demands more precise studies to reveal its resistome that plays a vital role in antibiotic resistance. In this study, we have identified and characterized a novel variant of CTX-M type ESBL i. e. CTX-M-152 in two *K. georgiana* isolated from WTPs located in Haryana, India. We have also reported a novel inhibitor of CTX-M-152 identified from a pool of natural products using computational approach such as high-throughput virtual screening, molecular docking and simulation studies along with some *in vitro* (MIC and ZOI) studies.

## 2. Materials and methods

### 2.1. Materials

Cefotaxime, and Ellagic acid were procured from Sigma-Aldrich (St. Louis, MO, USA). Antibiotic discs were purchased from Hi-media (Maharashtra, India). Other reagents and chemicals were of analytical grade.

### 2.2. Site of sample collection

A total of 102 water samples were collected from tertiary phase effluent spot of waste water treatment plant (Yamuna action sewage treatment plant) of Palwal region (Haryana, India). Pure cultures were obtained by inoculating the Luria Broth tubes with a single colony of each isolate.

### 2.3. Screening of ESBL producers

Clinical and Laboratory Standards Institute [CLSI, 2012] guidelines were followed to screen the ESBLproducers. Ceftazidime, Cefotaxime, and Ceftriaxone were used as indicator drugs having the same antibiotic concentration 30 μg each. These possible ESBL's producers were confirmed by phenotypic disc confirmatory test (PDCT) wherein a cefotaxime disc (CTX) with antibiotic concentration 30 μg alone and cefotaxime along with clavulanic acid i.e., cefotaxime-clavulanic acid (CEC) having concentration 20 + 10 μg were kept on the same plates from a distance of 30 mm between them. All the plates were incubated at 37 °C overnight and comparative zone of inhibition were interpreted. An ATCC (American Type Culture Collection) strains of *Klebsiella pneumoniae* (ATCC 700603) and *Escherichia coli* (ATCC 25922) were used as positive and negative controls respectively.

### 2.4. Amplification of *bla*$_{CTX\text{-}M}$ gene

The detection of *bla*$_{CTX\text{-}M}$ genes was performed with the help of *bla*$_{CTX\text{-}M}$ primers (forward primer 5′ATGGTTAAAAAATCACTGCG YCAGTTCACGC3′ and reverse primer 5′TTACAAACCGTYGGT GACGATTTTAGCCG3′) which gave an amplicon of ~860 bp [13]. Amplification of *bla*$_{CTX}$ gene was performed in total 50 μl of PCR reaction mixture, containing 3 μl DNA, 2 μl each of forward and reverse primers, 12 μl of 10× PCR buffer, 5 μl of MgCl2, 2 μl of 25 mM dNTP mix and 2 μl of Taq DNA polymerase (Genei, Bangalore, India). For *bla*$_{CTX\text{-}M}$ gene amplification, PCR was run with initial denaturation at 94 °C for 5 min followed by 35 cycles (denaturation at 94 °C for 45 s, annealing at 58 °C for 45 s and extension at 72 °C for 1 min) followed by final extension at 72 °C for 10 min [14]. PCR amplified products (5 μl) were loaded on 2% agarose gel and visualized under UV *trans*-illuminator (Genei, Banglore, India) and then photographed on Gel documentation system (BIO-RAD). Amplified gene was sequenced by SciGenome genomics service provider using 2nd set of *bla*$_{CTX\text{-}M}$ primers (forward primer 5′SCVATGTGCAGYACCAGTAA3′ and reverse primer 5′GCTGCCGGTYTTATCVCC3′) which gave an amplicon of ~513 bp.

### 2.5. Identification of bacterial strain on the basis of 16S rRNA gene

The detection of 16S rRNA genes were performed with the forward primer (~650bp) 5′GGCGGACGGGTGAGTAATG3′ and reverse primer 5′ATCCTGTTTGCTCCCCACG3′. Amplification of 16S rRNA gene was performed in 50 μl of PCR reaction mixture, containing 5 μl DNA, 2 μl each of downstream and upstream primers, 12 μl of 10× PCR buffer, 2 μl of 25 mM dNTP mix and 2 μl of Taq DNA polymerase. For 16S rRNA gene amplification, PCR was run with denaturation at 92 °C for 10 min followed by 35 cycles (denaturation at 94 °C for 1 min, annealing at 56 °C for 1 min and extension at 72 °C for 1 min) followed by final extension at 72 °C for 10 min. PCR products (5 μl) were loaded on 2% agarose gel and visualized under UV *trans*-illuminator (Genei, Banglore, India) and then photographed on Gel documentation system (BIO-RAD).

### 2.6. Generation of CTX-M-152 protein model

Molecular modeling was performed by submitting the primary sequence of CTX-M-152 to Swiss-Model server [15]. A search of suitable template has resulted in the identification of CTX-M-14 (PDB Id: 4ua6; X-ray resolution: 0.79 Å) as the best candidate with 0.97 GMQE (Global Model Quality Estimation) score and 88.89% identity.

### 2.7. Quality accessment of CTX-M-152 protein model

The quality of the generated model was accessed using Protein Structure Validation Suite (PSVS) ver. 1.5 available at http://psvs-1_5-dev.nesg.org/ [16]. It describes all the validation parameters such as Procheck, Verify3D, ProsaII, RMSDs, Molprobity clash score and Ramachandran plot.

### 2.8. Preparation of protein

The CTX-M-152 protein was prepared for molecular docking using protein preparation wizard of Schrodinger suite (Schrodinger, LLC, NY, USA) as described previously [17,18]. Briefly, missing hydrogen atoms were added, bond orders were assigned and any heterogeneous atoms were deleted. Further, water molecules forming less than two contacts either with protein or ligand were deleted. Furthermore, any missing side chains or loops were added using Prime (Schrodinger, LLC, NY, USA). Finally, the CTX-M-152 protein was optimized to create hydrogen bond network and the energy of the whole system was minimized using OPLS3e (Optimized Potentials for Liquid Simulations) forcefield. The most probable binding site of CTX-M-152 was identified using SiteMap (Schrodinger, LLC, NY, USA) and the grid box (80 × 80 × 80 Å) was created by selecting the active site amino acid residues in receptor-grid generation module of Maestro (Schrodinger, LLC, NY, USA).

### 2.9. Preparation of ligands

Novel inhibitors of CTX-M-152 were identified by screening a library (L1400) of natural compounds available at www.selleckchem.com using Glide (Schrodinger, LLC, NY, USA) accessed on Sep 16, 2018. The L1400 library contains 803 unique, structurally diverse, bioactive and cell permeable compounds/ligands in structure data file (sdf) format. These ligands were prepared in LigPrep (Schrodinger, LLC, NY, USA) before molecular docking by generating different ionization states at pH 7.0 ± 2.0 using Epik (Schrodinger, LLC, NY, USA) followed by desaltation. A maximum of 32 conformers were generated for a particular ligand. Finally, the energy of these ligands were minimized using OPLS3e forcefield.

### 2.10. Molecular docking

The potential of natural compounds to bind CTX-M-152 at the active site was evaluated by performing molecular docking in three stages, namely, HTVS (High Throughput Virtual Screening) followed by SP (Standard Precision) and XP (Extra Precision) using Glide (Schrodinger, LLC, NY, USA) as reported earlier [18]. XP docking was performed by keeping scaling factor and partial charge cut off to the default values of 0.80 and 0.15 respectively. The docking results were analyzed and protein-ligand interaction figures were prepared in Maestro (Schrodinger, LLC, NY, USA). The docking affinity ($K_b$) of compounds for the active site of CTX-M-152 was calculated from docking energy ($\Delta G$) using the following relation as reported previously [19,20].

$$\Delta G = -RT \ln K_b \tag{1}$$

where, $R$ and $T$ were Boltzman gas constant (= 1.987 cal mol$^{-1}$ K$^{-1}$) and temperature (= 298 K) respectively.

### 2.11. MM-GBSA (molecular mechanics-generalized born surface area) calculations

The effect of solvent on the binding affinity of compounds towards CTX-M-152 was evaluated by performing MM-GBSA using MM forcefields along with implicit solvation model as described earlier [17]. MM-GBSA calculations integrates OPLS3e forcefield, VSGB solvent model and rotamer search algorithms [21]. Prime (Schrodinger, LLC, NY, USA) with default setting wherein protein atoms were considered rigid while ligand atoms were relaxed was used to calculate MM-GBSA of protein-ligand systems having an XP score $\leq -6.0$ kcal mol$^{-1}$ (5 compounds) on the basis of the following equation.

$$\Delta G = \left[E_{complex} - \left(E_{ligand} + E_{receptor}\right)\right] \tag{2}$$

### 2.12. Molecular dynamics simulation

Molecular dynamics simulation was performed to check the stability of CTX-M-152-inhibitor complex using Desmond (Schrodinger, LLC, NY, USA) [22]. An orthorhombic simulation box was placed in such a way as to keep the boundaries of the box at least 10 Å away from the CTX-M-152-inhibitor complex. TIP3P explicit water model was used to solvate the simulation box and proper counterions were also added to neutralize the system. Further, 150 mM NaCl was also added to mimic the physiological condition. The whole system was minimized with 1000 iteration with convergence criteria of 1 kcal mol$^{-1}$ Å$^{-1}$ with the help of OPL3e forcefield. The simulation was run for 50 ns using NTP ensemble at 298 K temperature and 1.013 bar pressure. Nose-Hoover Chain thermostat and Matrtyna-Tobias-Klein barostat were used to maintain the temperature and pressure respectively [23,24]. Energies and structures were recorded every 10 ps interval and a time step of 2 fs was maintained during the simulation process.

### 2.13. Physiochemical and ADMET properties

The physiochemical properties of the selected compounds were determined using PubChem database (www.pubchem.ncbi.nlm.nih.gov). Moreover, the ADMET (Adsorption, Distribution, Metabolism, Excretion and Toxicology) properties of the selected compounds were determined using QikProp (Schrodinger, LLC, NY, USA) to evaluate the drug-likeliness of the compounds.

### 2.14. In vitro studies of Ellagic acid on Kluyvera georgiana

The antibacterial activities of Ellagic acid against *Kluyvera georgiana* isolates were performed by disc-diffusion method. In brief, Overnight grown bacterial suspension was spread on Mueller Hinton Agar (MHA) plates and Ellagic acid was applied using sterilized cork borer. A disc of cefotaxime (30 μg) was used as positive control on same plate for both *Kluyvera georgiana* isolates. After incubation, effectiveness of Ellagic acid was performed by measuring the zone of inhibition. The minimum inhibitory concentration (MIC) was performed by broth micro dilution method using 96 well plate.

## 3. Results

### 3.1. Screening of ESBL producers by Kirby-Bauer disc diffusion method

Screening of ESBL's producers was performed by plating pure cultures in the presence of third generation Cephalosporins. The zone of inhibition diameter of ≤22 mm, ≤27 mm, ≤25 mm and ≤27 mm for Ceftazidime, Cefotaxime, Ceftriaxone and Aztreonam respectively were suspected of ESBL producers (Table 1). Here, we have screened 102 non-duplicate bacterial isolates from the Yamuna sewage treatment plant located in Palwal region of Haryana, India. The screening of bacterial isolates by Kirby-Bauer disc diffusion method against third generation of Cephalosporins (i.e. Cefotaxime, Ceftazidime, and Ceftriaxone) has led

**Table 1**
Antibiotic resistance profile of ESBL positive bacterial isolates.

| Bacterial Isolates | AT | CTR | CAZ | CTX | CAC | CEC |
|---|---|---|---|---|---|---|
| AT-4 | 11 | 13 | 0 | 10 | 8 | 19 |
| AT-7 | 8 | 11 | 7 | 16 | 15 | 23 |
| AT-12 | 12 | 0 | 0 | 19 | 8 | 23 |
| AT-15 | 11 | 12 | 0 | 16 | 8 | 22 |
| AT-19 | 2 | 8 | 0 | 24 | 9 | 31 |
| AT-23 | 17 | 2 | 23 | 22 | 30 | 30 |
| AT-24 | 8 | 10 | 0 | 0 | 8 | 9 |
| AT-26 | 11 | 6 | 0 | 9 | 9 | 16 |
| AT-28 | 12 | 8 | 20 | 13 | 27 | 20 |
| AT-29 | 6 | 8 | 4 | 14 | 10 | 20 |
| AT-30 | 7 | 6 | 6 | 17 | 14 | 24 |
| EC | n.d. | 20 | 25 | 22 | n.d. | n.d. |
| KP | n.d. | 09 | 21 | 16 | n.d. | n.d. |

Concentration of antibiotics were AT (Aztreonam - 30 μg), CTR (Ceftriaxone - 30 μg), CTX (Cefotaxime - 30 μg), CAZ (Ceftazidime - 30 μg), CAC (Ceftazidime + Clavulanate - 20 + 10 μg), CEC (Cefotaxime + Clavulanate - 20 + 10 μg). EC stands for *Escherichia coli* ATCC 25922 and *Klebsiella pneumoniae* (ATCC 700603). Zone of inhibition (ZOI) diameter was measured in millimetres (mm). n.d. stands for not determined.

to the identification of 20 (19.6%) ESBLs producers.

### 3.2. Phenotypic disc confirmatory test (PDCT)

The identity of 20 ESBL producers was again confirmed by PDCT following Clinical and Laboratory Standards Institute (CLSI, 2012) guidelines. The zone of inhibition diameter of CAC was compared with antibiotic CAZ alone. Similarly, the zone of inhibition diameter of CEC was compared with the antibiotic CTX alone. The screened isolates showing ≥5 mm increase in the zone of inhibition diameter when tested with antibiotic + clavulanic acid than the antibiotic tested alone clearly indicated the presence of ESBL producers. Out of 20 isolates, 11 bacterial isolates were confirmed to produce ESBLs on the basis of their double sensitivity towards CAC as well as CAZ (Table 1).

### 3.3. Amplification of $bla_{CTX-M}$ gene by PCR

The screening of ESBL positive isolates for the presence of $bla_{CTX-M}$ was conducted by PCR. An amplicon of ~860 bp fragment was observed in all the 11 bacterial isolates i.e. 55% (Fig. 1).

### 3.4. Identification of bacterial species based on 16S rRNA gene

The nature of bacterial species harboring $bla_{CTX-M}$ gene was identified by performing PCR amplification of 16S rRNA gene. Keeping into consideration the information of variable and conserved domains of the 16S rRNA gene, designed primers have led to the amplification of ~650 bp fragment (Supplementary Fig. S1). The sequencing of PCR amplicon at SciGenome genomics services and its analysis by submitting the sequences to NCBI has led to the identification of 11 $bla_{CTX-M}$ positive isolates distributed as one strain of *Escherichia fergusonii* (9.1%), two strains of *Escherichia coli* (18.2%), two strains of *Shigella* sp. (18.2%), four strains of *Kluyvera* sp. (36.4%), and two strains of *Enterobacter* sp. (18.2%) (Table 2). Of the 4 isolates of *Kluyvera* sp., 3 had clear identity of being *Kluyvera georgiana* (GenBank accession numbers 16S rRNA: KY855399, KY855400 & KY855401). Of them, only two isolates were found to harbor $bla_{CTX-M-152}$ gene, which encodes β-lactamase CTX-M-152 protein (GenBank accession numbers: KY860788 and KY860789) with similar primary amino acid sequence (Table 2).

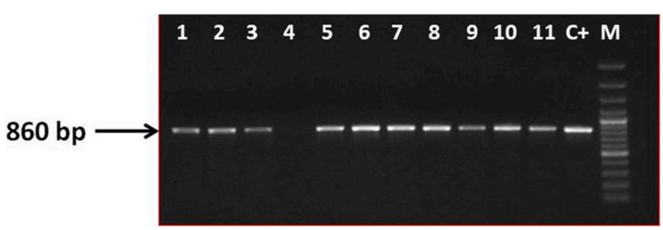



**Fig. 1.** PCR amplification of $bla_{CTX-M}$gene (~860 bp) in 11 ESBL positive bacterial strains. Lane M: 100 bp marker, Lanes 1–11: amplified $bla_{CTX-M}$ gene, Lane C+: positive control.

### 3.5. Structure prediction by Swiss-Model and its quality accessment

The identification of an inhibitor of a protein or an enzyme using computational approaches is an emerging tool in the endeavor of drug discovery. Here, the three-dimensional structure of CTX-M-152 model protein was generated by submitting its amino acid sequence to Swiss-Model server available at https://swissmodel.expasy.org/ [25]. CTX-M-14 (PDB Id: 4ua6; X-ray resolution: 0.79 Å) was identified as the suitable template (88.89% identity) to build the model structure of CTX-M-152. Fig. 2A shows an alignment of template (CTX-M-14; 4ua6) and target (CTX-M-152) sequences. The amino acid residues critical for the hydrolysis of β-lactam antibiotics have been shown in red color. Fig. 2B shows the three-dimensional structure of the generated model of CTX-M-152, wherein all the significant amino acid residues were conserved. Fig. 2C shows the Ramachandran plot (Procheck) in which all the amino acid residues occupied either the most favored (91.9%) or generously and additionally allowed (8.1%) regions (Supplementary Table S1). Similarly, the Ramachandran plot scores according to Richardson's lab showed that the modelled CTX-M-152 protein had occupied 97.0% of the most favored region and 3.0% in the generously and additionally allowed regions (Supplementary Table S1). The Z-score of the generated model also indicated that the overall quality of the model was appropriate for further studies (Fig. 2D). The overall quality of CTX-M-152 model protein was accessed by parameters such as Verify3D, ProsaII, Procheck, RMSDs and Molprobity clash score using protein structure validation suite (PSVS) and the results are presented in (Supplementary Table S2).

### 3.6. Analysis of HTVS, SP and XP docking

HTVS was performed on a library of natural compounds (L1400) from Selleck which had 803 unique and structurally diverse compounds. A total of 347 compounds with variable affinities towards the active site of CTX-M-152 model protein were identified (Supplementary Table S3). Out of 347 compounds, 44 with HTVS docking score of ≤ −5.0 kcal $mol^{-1}$ (~top 1%) were selected for SP docking. Out of 44 compounds, only 34 showed an affinity towards CTX-M-152 active site in SP docking (Table 3). The compounds were further filtered by performing XP docking with SP docking score of ≤ −5.0 kcal $mol^{-1}$ (25 compounds). The top 5 compounds (≤−6.0 kcal $mol^{-1}$) on the basis of XP score were identified as Naringin dihydrochalcone, Salvianolic acid B, Inositol, Guanosine and Ellagic acid (Table 4). These 5 compounds were further evaluated on the basis of physiochemical and MM-GBSA calculations along with ADMET properties to identify the most potent natural compound inhibitor of CTX-M-152.

**Table 2**
Identification of bacterial species by 16S rRNA sequencing.

| Bacterial Isolates | GenBank accession number (CTX-M) | CTX-M Variant | GenBank accession number (16S rRNA) | 16S rRNA Species |
|---|---|---|---|---|
| AT-4 | KY860785 | CTX-M-152 | KY855393 | *Escherichia fergusonii* |
| AT-7 | KY860782 | CTX-M-15 | KY855394 | *Shigella* sp. |
| AT-12 | KY860783 | CTX-M-15 | KY855395 | *Shigella* sp. |
| AT-15 | KY860786 | CTX-M-152 | KY855396 | *Enterobactor cloacae* |
| AT-19 | n.a. | n.a. | KY855397 | *Kluyvera* sp. |
| AT-23 | KY860784 | CTX-M-15 | KY855398 | *Enterobactor* sp. |
| AT-24 | n.a. | n.a. | KY855399 | *Kluyvera georgiana* |
| AT-26 | KY860787 | CTX-M-152 | n.a. | *Escherichia coli* |
| AT-28 | KY860788 | CTX-M-152 | KY855400 | *Kluyvera georgiana* |
| AT-29 | KY860789 | CTX-M-152 | KY855401 | *Kluyvera georgiana* |
| AT-30 | KY860790 | CTX-M-152 | KY855402 | *Escherichia coli* |

n.a. represents not allocated.

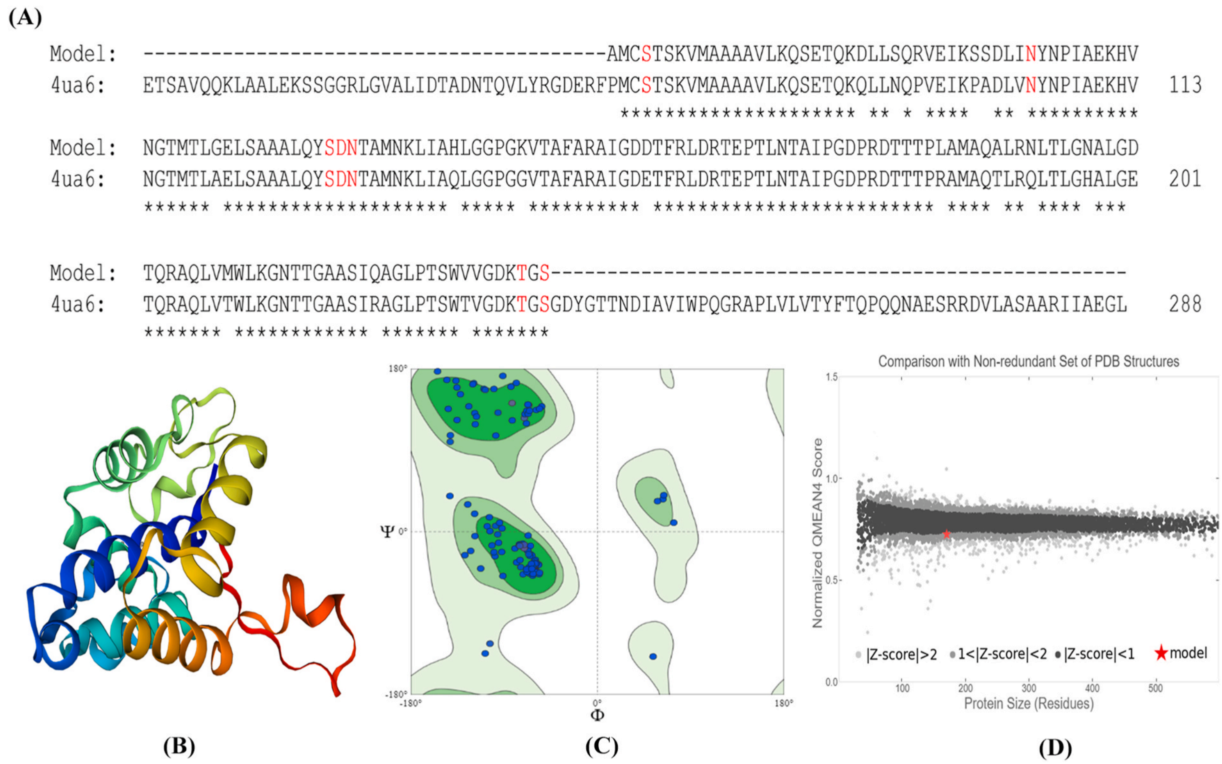



**Fig. 2.** Structural and functional analysis of CTX-M-152. (A) Alignment of CTX-M-152 (target) and CTX-M-14 (PDB ID: 4UA6; template) sequences. The residues shown in red color are critically involved in the hydrolysis of β-lactam antibiotics, (B) Three-dimensional structure of the generated CTX-M-152 model, (C) Ramachandran plot of CTX-M-152 model showing that all the residues fall into allowed and/or generously allowed regions, and (D) ProsaII profile of CTX-M-152 model showing Z-score.

### 3.7. Physiochemical properties and MM-GBSA calculations

Physiochemical properties of the selected compounds (Naringin dihydrochalcone, Salvianolic acid B, Inositol, Guanosine and Ellagic acid) were documented from PubChem database (www.pubchem.ncbi.nlm.nih.gov). The results presented in Table 5 revealed that the physiochemical properties of Inositol, Guanosine and Ellagic acid obeyed Lipinski's rule of five, whereas the molecular weights of Naringin dihydrochalcone and Salvianolic acid were >500 g/mol. The effect of solvent on the binding affinity of shortlisted compounds (Naringin dihydrochalcone, Salvianolic acid B, Inositol, Guanosine and Ellagic acid) towards CTX-M-152 was determined by MM-GBSA calculations (Table 5). The MM-GBSA analysis has shown that Ellagic acid had the lowest MM-GBSA energy (−49.55 kcal mol$^{-1}$) followed by Naringin dihydrochalcone (−46.38 kcal mol$^{-1}$), Salvianolic acid B (−45.42 kcal mol$^{-1}$), Guanosine (−35.14 kcal mol$^{-1}$) and Inositol (−33.34 kcal mol$^{-1}$). It is clear that the MM-GBSA energy of Ellagic acid was the lowest, suggesting a strong interaction between Ellagic acid and CTX-M-152. The MM-GBSA of Naringin dihydrochalcone and Salvianolic acid were also close to that of Ellagic acid, but these were not considered for further evaluation owing to their deviation from Lipinski's rule of five (molecular weight > 500 g/mol).

### 3.8. Validation of screening protocol by ROC (receiver operating characteristic) analysis

Authenticity of the virtual screening protocol adopted in this study was confirmed by performing ROC analysis. It is a graphical representation of a relation between the sensitivity and specificity of the screening protocol. We had employed Schrodinger's Enrichment calculator (Schrodinger, LLC, NY, USA) to check whether the adopted screening protocol can detect the active compounds from a pool of inactive decoys and place them in the top % of a ranked database. Here, we had used Schrodinger's standard decoy set for the enrichment calculation by computing area under curve [26,27]. The top 5 compounds identified after MM-GBSA were subjected to enrichment calculations. The ROC value and area under curve obtained for MM-GBSA short listed compounds was 0.84 and 0.84 respectively (Fig. 3).

### 3.9. Molecular docking and molecular dynamics simulation

Interaction between Ellagic acid and CTX-M-152 was evaluated by performing molecular docking and the results are presented in Fig. 4A and B and Table 6. Moreover, molecular docking between Cefotaxime (control ligand) and CTX-M-152 was also performed for comparative analysis (Fig. 4C and D and Table 6). The results demonstrate that Ellagic acid was bound comfortably at the active site of CTX-M-152 and the complex was stabilized by six hydrogen bonds (Ser70, Asn104, Asn132, Asn170, Thr235, and Ser237), three carbon-hydrogen bonds (Ser70 and Thr216), one Pi-donor hydrogen bond (Asn132) and three hydrophobic interactions (Pi-Pi stacked) with Tyr105 (Fig. 4A and B and Table 6). In addition, a single Pi-lone pair interaction was also formed between Ellagic acid and Ser237 of CTX-M-152. Other amino acid residues that participated in stabilizing the Ellagic acid-CTX-M-152 complex were Cys69, Ser130, Glu166, Lys234, and Gly236. The whole complex was stabilized by a binding energy of −6.665 kcal mol$^{-1}$ which corresponded to the docking affinity of 7.73 × $10^4$ M$^{-1}$ (Table 6). As a comparison, the molecular docking of Cefotaxime (control) with CTX-M-152 was also performed. We found that CTX-M-152-cefotaxime complex formation was driven by the formation of nine hydrogen bonds (Ser70, Asn104, Asn132, Asn170, Thr235, Lys234, and Ser237), four carbon-hydrogen bond (Ser70, Pro167, and Thr216), two electrostatic interactions (Lys234 and Ser237), and one Pi-alkyl hydrophobic interaction (Tyr105) (Fig. 4C and D and Table 5). The whole CTX-M-152-Cefotaxime complex was stabilized by −7.022 kcal mol$^{-1}$ of free energy which corresponds to the binding affinity of 1.41 × $10^5$ M$^{-1}$

**Table 3**
SP docking of selected compounds on the basis of HTVS score (≤−5.0 kcal mol$^{-1}$) against CTX-M-152 model protein.

| S. No. | Compound name | Glide g-score (kcal mol$^{-1}$) | Glide energy (kcal mol$^{-1}$) | Glide e-model (kcal mol$^{-1}$) |
|---|---|---|---|---|
| 1. | Naringin dihydrochalcone | −6.535 | −54.364 | −72.595 |
| 2. | Inositol | −6.111 | −28.525 | −38.372 |
| 3. | Guanosine | −6.047 | −35.242 | −49.358 |
| 4. | Salvianolic acid B | −6.034 | −57.069 | −73.415 |
| 5. | Hesperetin | −5.853 | −31.915 | −43.045 |
| 6. | 4′-Methoxyresveratrol | −5.727 | −30.920 | −40.118 |
| 7. | Daphnetin | −5.707 | −29.546 | −40.623 |
| 8. | Ellagic acid | −5.736 | −37.583 | −50.908 |
| 9. | 2,6-Dihydroxypurine | −5.785 | −25.442 | −33.913 |
| 10. | Brefeldin A | −5.451 | −30.045 | −39.595 |
| 11. | Shikimic acid | −5.400 | −25.904 | −34.898 |
| 12. | Inosine | −5.393 | −35.027 | −45.705 |
| 13. | Gallic acid | −5.339 | −25.767 | −35.032 |
| 14. | 4-Methylumbelliferone | −5.336 | −24.669 | −33.069 |
| 15. | Quinolinic acid | −5.331 | −20.395 | −28.355 |
| 16. | Dihydrothymine | −5.296 | −22.290 | −29.369 |
| 17. | Sesamol | −5.294 | −21.875 | −29.935 |
| 18. | Umbelliferone | −5.278 | −24.075 | −32.426 |
| 19. | Serotonin | −5.247 | −23.064 | −34.265 |
| 20. | Dopamine | −5.158 | −25.426 | −33.793 |
| 21. | Glucosamine | −5.280 | −25.303 | −33.738 |
| 22. | Asaraldehyde | −5.049 | −23.994 | −30.606 |
| 23. | Esculetin | −5.056 | −25.080 | −34.225 |
| 24. | Vanillin | −5.094 | −21.465 | −28.300 |
| 25. | Scoparone | −5.004 | −24.238 | −32.303 |
| 26. | Tetrahydropapaverine | −4.988 | −30.889 | −40.606 |
| 27. | Resveratrol | −4.917 | −26.514 | −34.592 |
| 28. | Bengenin | −4.899 | −33.251 | −42.404 |
| 29. | (−)Epicatechin | −4.692 | −33.993 | −39.941 |
| 30. | Paeonol | −4.830 | −20.032 | −27.167 |
| 31. | Caffeic acid | −4.606 | −26.739 | −33.446 |
| 32. | Oxyresveratrol | −4.590 | −27.839 | −37.098 |
| 33. | Rheochrysidin | −4.228 | −22.908 | −22.035 |
| 34. | Nordihydroguaiaretic acid | −3.709 | −28.815 | −32.018 |

**Table 4**
XP docking of selected compounds having ≤ −5.0 kcal mol$^{-1}$ of SP score against CTX-M-152 model protein.

| S. No. | Compound name | Glide g- score (kcal mol$^{-1}$) | Glide energy (kcal mol$^{-1}$) | Glide e-model (kcal mol$^{-1}$) |
|---|---|---|---|---|
| 1. | Naringin dihydrochalcone | −8.160 | −49.113 | −71.252 |
| 2. | Salvianolic acid B | −7.209 | −59.754 | −80.052 |
| 3. | Inositol | −6.831 | −27.328 | −31.412 |
| 4. | Guanosine | −6.762 | −33.68 | −42.537 |
| 5. | Ellagic acid | −6.665 | −37.594 | −48.038 |
| 6. | Shikimic acid | −5.682 | −24.121 | −30.123 |
| 7. | Glucosamine | −5.776 | −23.338 | −26.409 |
| 8. | Daphnetin | −5.579 | −28.013 | −36.89 |
| 9. | Dopamine | −5.497 | −22.915 | −25.558 |
| 10. | Inosine | −5.222 | −30.742 | −37.17 |
| 11. | Gallic acid | −4.871 | −26.266 | −31.121 |
| 12. | Brefeldin A | −4.549 | −29.063 | −38.195 |
| 13. | 4′-Methoxyresveratrol | −4.427 | −25.549 | −32.44 |
| 14. | Hesperetin | −4.349 | −33.886 | −38.57 |
| 15. | Esculetin | −4.048 | −25.092 | −30.968 |
| 16. | Serotonin | −3.965 | −25.555 | −25.03 |
| 17. | Asaraldehyde | −3.885 | −24.189 | −25.614 |
| 18. | 2,6-Dihydroxypurine | −4.061 | −24.608 | −31.273 |
| 19. | Vanillin | −3.868 | −21.467 | −25.195 |
| 20. | 4-Methylumbelliferone | −3.709 | −24.283 | −30.434 |
| 21. | Scoparone | −3.637 | −24.594 | −28.647 |
| 22. | Quinolinic acid | −3.523 | −17.395 | −18.404 |
| 23. | Dihydrothymine | −3.497 | −20.899 | −25.986 |
| 24. | Umbelliferone | −3.476 | −24.359 | −29.994 |
| 25. | Sesamol | −3.180 | −20.56 | −23.944 |

(Table 6). It is worth to notice that both Ellagic acid and Cefotaxime had occupied amino acid residues critical for the hydrolytic activity of CTX-M-152. The residues such as Ser70, Asn104, Tyr105, Asn132, Asn170, Thr216, Thr235 and Ser237 were commonly involved in the interaction with Ellagic acid as well as Cefotaxime. It is worth to note that these residues have been reported to play a crucial role in maintaining the active site of CTX-M-152 and hydrolysis of β-lactam antibiotics.

The stability of interaction between CTX-M-152 and Ellagic acid was accessed by performing molecular dynamics simulation for 50 ns under physiological conditions. Fig. 5A shows RMSDs of CTX-M-152 alone and in the presence of Ellagic acid with respect to the initial conformation. Fig. 5B signifies the change in molecular surface area (MolSA), solvent accessible surface area (SASA) and polar surface area (PSA) of CTX-M-152 as a function of simulation time. It is clear that the RMSD value of CTX-M-152 alone initially fluctuated between 0 and 4 Å within the first 10 ns and then gets stabilized thereafter. Similarly, the RMSD value of the CTX-M-152-Ellagic acid complex fluctuated between 1.9 Å to 9.5 Å during the first 8 ns and then gets stabilized after 10 ns (Fig. 5A). The initial fluctuations in the RMSD values can be attributed to the entry of Ellagic acid into the active site of CTX-M-152. Further, the changes in molecular surface area (MolSA), solvent accessible surface area (SASA) and polar surface area (PSA) of CTX-M-152 as a function of simulation time has clearly shown that MolSA and PSA were constant throughout the simulation time, while SASA became constant only after initial fluctuations for the first 15 ns. Taken together, these results confirm that the interaction between CTX-M-152 and Ellagic acid was stable in nature.

### 3.10. ADMET properties

ADMET properties of Ellagic acid was determined to ensure that it possess drug-like features. The values of QPpolrz (polarizability), QPlogS (aqueous solubility), QPlogPC16 (hexane/gas), QPlogPoct (octanol/gas), QPlogPw (water/gas), QPlogPo/w (octanol/water), QPlogKp(skin permeability) and QPlogKhsa (serum protein binding) for Ellagic acid were estimated to be 23.52, −1.97, 9.62, 20.36, 16.82, −1.34, −6.78 and −0.68 respectively (Supplementary Table S4). The analysis of ADMET properties has shown that all the descriptors were within the limits of a drug-like molecule.

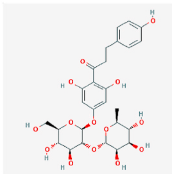
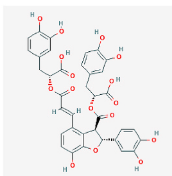
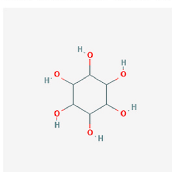
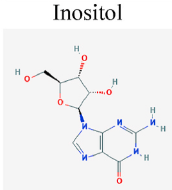
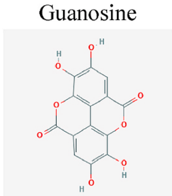
### 3.11. Antibacterial activity of Ellagic acid

The Ellagic acid dissolved in 10% DMSO was used for the studies. The minimum inhibitory concentration (MIC) of Ellagic acid (0.58–600 μg) and Cefotaxime (2–1024 μg) by broth micro dilution method revealed MIC value of 300 μg for Ellagic acid for both strains of *Kluyvera georgiana* and >1024 μg of Cefotaxime for AT-28 and > 64 μg of Cefotaxime for AT-29 *Kluyvera georgiana* isolates, respectively. The results represent mean value of 3 readings after subtracting the factional effect attributed by DMSO. To strengthen our results, the experiments was supported by antimicrobial activity assay of Ellagic acid along with Cefotaxime. Study of the zone of inhibition in the antimicrobial activity experiment for Ellagic acid further strengthen our analysis as its ZOI was 17 mm and 18 mm for AT-28 and AT-29 *Kluyvera georgiana* isolates that are comparative to ZOI of Cefotaxime (20 mm and 18 mm for AT-28 and AT-29 respectively) (Supplementary Fig. S2).

## 4. Discussion

Aquatic environments such as lakes, rivers, streams, ponds, etc serve as a leading site for the development of microbial resistance. Aquatic environments with effluents from anthropogenic sources such as hospital, municipal and aquaculture contain critical levels of antimicrobials, pesticides, fungicides and some other toxic compounds which progressively led to development of resistance among microbial inhabitants [28,29]. In China and the United States, tetracycline resistant

**Table 5**
Physicochemical properties of selected compounds (XP score ≤ −6.0 kcal $mol^{-1}$) using PubChem database.

| S. No. | Name and structure of compound | MW (g/mol) | HBD | HBA | Tpsa ($Å^2$) | RB | NC | MM-GBSA (kcal $mol^{-1}$) |
|---|---|---|---|---|---|---|---|---|
| 1. |  Naringin dihydrochalcone | 582.5 | 9 | 18 | 236 | 9 | 0 | −46.38 |
| 2. |  Salvianolic acid B | 718.6 | 9 | 16 | 278 | 14 | 0 | −45.42 |
| 3. |  Inositol | 180.2 | 6 | 10 | 121 | 0 | 0 | −33.34 |
| 4. |  Guanosine | 283.2 | 5 | 7 | 155 | 2 | 0 | −35.14 |
| 5. |  Ellagic acid | 302.2 | 4 | 8 | 134 | 0 | 0 | −49.55 |

MW: Molecular weight; HBD: Hydrogen bond donor; HBA: Hydrogen bond acceptor; Tpsa: Total polar surface area; RB: Rotatable bonds; NC: Net charge; MM-GBSA: Molecular Mechanics-General Born Surface Area.

genes, ribosomal protection proteins (RPP), efflux protein (EFP) gene have been detected in bacteria isolated from the discharge of sewage treatment plants (STPs) and feed lot lagoons [28,30]. Similarly, sulfonamides which was used in prophylaxis against bacteria have been found in wastewater treatment plants (WWTPs), sludge and animal additives [30,31]. ESBL-producing *Klebsiella pneumonia* was found harboring resistant genes against aminoglycoside in the effluent of STP and activated sludge, and are believed to enter different ecosystem through food cycle [28]. In India, increasing load of MDR bacteria have been reported from major rivers such as Ganges, Yamuna and Cauvery [32–35]. The major sources of river contamination are the discharges from pharmaceutical industries and hospitals without any adequate treatment [36,37].

In this study, isolation and screening of water samples for bacterial isolates by Kirby-Bauer disc diffusion method against third generations of Cephalosporins (i.e. Cefotaxime, Ceftazidine and Ceftriaxone), has led to the identification of 20 (19.6%) ESBLs producers (Table 1). The presence of ESBLs was further confirmed by phenotypic disc confirmatory test (PDTC). The $bla_{\text{CTX-M}}$ gene in ESBLs positive bacterial isolates was amplified using PCR, sequenced and the sequence was submitted to GenBank. The analysis of results revealed that 11 ESBL positive bacterial isolates harbor $bla_{\text{CTX-M}}$ genes. Similarly, the 16S rRNA gene of 11 bacterial isolates containing $bla_{\text{CTX-M}}$ gene was amplified, sequenced and the sequence was submitted to GenBank. The analysis of GenBank results showed that 2 *Kluyvera georgiana* isolates were positive for $bla_{\text{CTX-M-152}}$ gene, encoding CTX-M-152 protein with similar amino acid sequence. Since there is no report of the three-dimensional structure of CTX-M-152, we modelled the structure of CTX-M-152 and used it for the screening of a potential inhibitors. In a previous report, the phylogenetic analysis of CTX-M-152 showed that it belongs to CTX-M-25 group [13]. A comparison of amino acid sequences of CTX-M-152 and CTX-M-25 revealed that CTX-M-152 possesses 10 point mutations such as Ile7-Met, Tyr24Cys, His26Gln, Ala77Val, Asp89Gly, Ser99Gly, Leu119Phe, Gln146Asp, Ala154Thr, and Asp240Gly.

The catalytic site of CTX-M-152 is characterized by the presence of four conserved motifs namely Ser70-Thr71-Ser72-Lys73 motif (forms roof of the active site), Ser130-Asp131-Asn132 motif (located on a short loop in the α-domain and forms right side of the active site), Glu166-Pro167-Thr168-Leu169-Asn170 motif (located on the Ω loop and forms floor of the active site), and Lys234-Thr235-Gly236 motif (located on the β3 strand in the αβ domain and forms left side of the active site). Molecular docking of Ellagic acid at the catalytic site of CTX-M-152 has

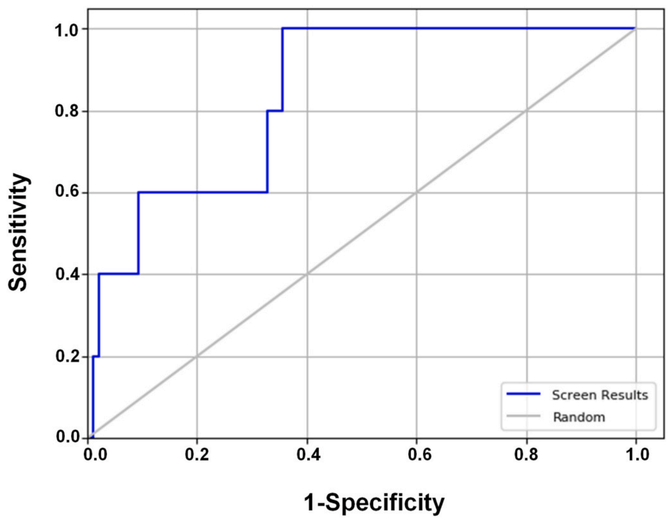



**Fig. 3.** Validation of the screening protocol by ROC-AUC analysis. It represents the relationship between sensitivity and specificity of XP docking.

shown that it form hydrogen bonds and hydrophobic interactions with the amino acid residues such as Ser70, Asn104, Asn132, Asn170, Thr216, Thr235, and Ser237 belonging to different conserved motifs of CTX-M-152. Among them, the residues Ser70, Asn132, and Ser237 are active site residues, involved in the hydrolysis of β-lactam substrates. The amino acid residue Ser237 plays a significant role in facilitating the formation of a productive enzyme-substrate complex. It forms a hydrogen bond with the carboxylate oxygen of substrate, which in turn brings the carbonyl group of the substrate to the proximity of oxyanion hole, and hence promote the hydrolysis of the substrate [32]. The active site residue Ser70, a part of a conserved Ser70-xxx-xxx-Lys73 sequence, along with Glu166 and a catalytic water molecule directly participate in the hydrolysis of β-lactam antibiotics through an acid-base catalytic mechanism [38]. Another residue Asn104 along with Tyr105 form a bend in the active site of CTX-M-9 and participate in forming a hydrogen bond with the carbonyl group of substrates [39]. Similarly, Asn132 is a part of a conserved motif Ser130-Asp131-Asn132 which forms a part of the active site cavity.

The identification of an inhibitor of a protein or an enzyme using computational approaches is an emerging tool in the endeavour of drug discovery. Here, we performed HTVS of natural compounds library (L1400) from Selleck and screened most potent compounds ($\leq -6.0$ kcal mol$^{-1}$) for their binding against the three-dimensional model of CTX-M-

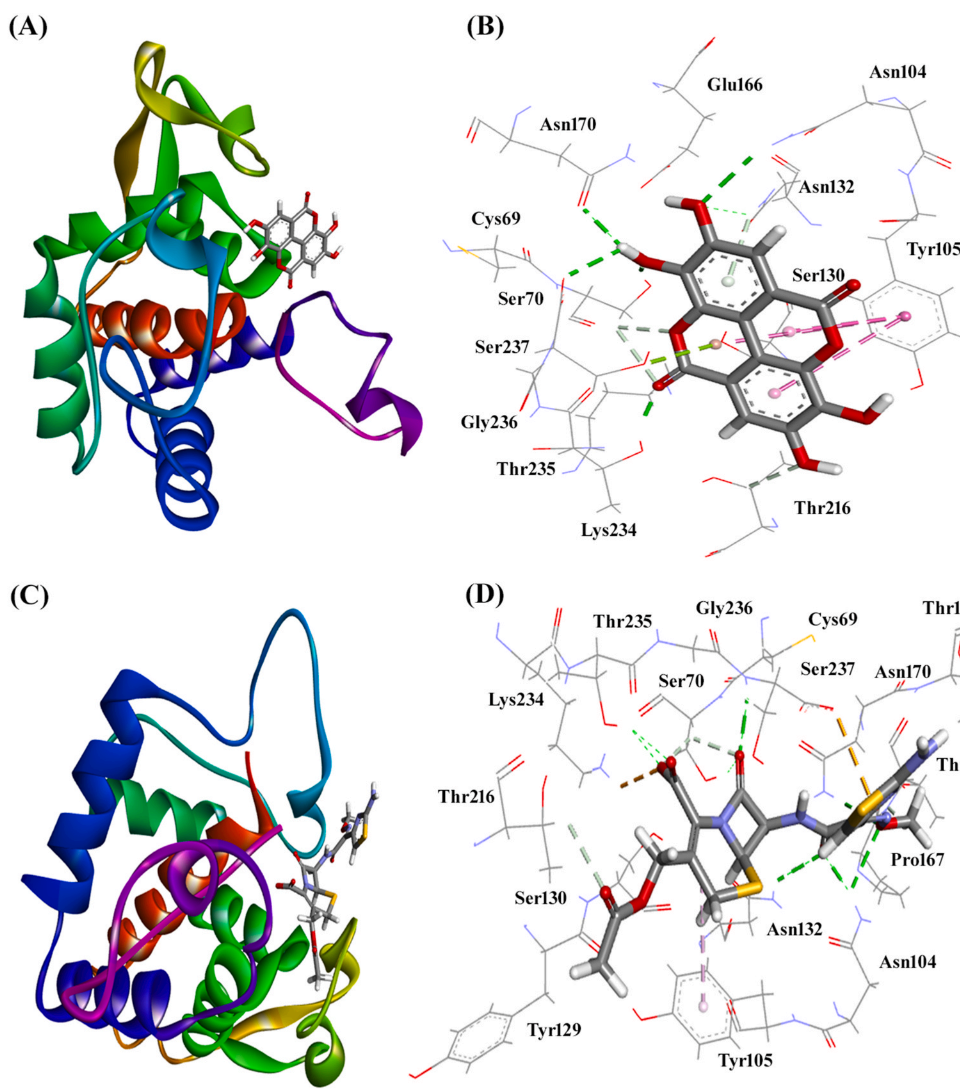



**Fig. 4.** (A) Molecular docking of Ellagic acid at the active site of CTX-M-152, and (B) Nature of interaction between Ellagic acid and CTX-M-152, (C) Molecular docking of Cefotaxime (control) at the active site of CTX-M-152, and (D) Nature of interaction between Cefotaxime and CTX-M-152.

**Table 6**
Docking and interaction parameters between CTX-M-152 model protein with Ellagic acid and Cefotaxime.

| Nature of interaction | Type of Interaction | Distance (Å) | Docking energy (kcal $mol^{-1}$) | Docking affinity ($M^{-1}$) |
|---|---|---|---|---|
| **Ellagic acid (EA)** | | | | |
| Ser70:HG - EA:O | Hydrogen Bond | 1.9496 | −6.665 | $7.73 \times 10^4$ |
| Asn104: HD22 - EA: O | Hydrogen Bond | 2.4344 | | |
| Asn132: HD22 - EA | Hydrogen Bond | 2.4470 | | |
| Thr235:HG1 - EA:O | Hydrogen Bond | 1.9567 | | |
| EA:H - Asn170: OD1 | Hydrogen Bond | 2.1625 | | |
| EA:H - Ser237:O | Hydrogen Bond | 2.4460 | | |
| Ser70:HB2 - EA:O | Carbon Hydrogen Bond | 2.7219 | | |
| Ser70:HB2 - EA:O | Carbon Hydrogen Bond | 2.9360 | | |
| Thr216:HB - EA:O | Carbon Hydrogen Bond | 2.7312 | | |
| Asn132: HD22 - EA | Hydrogen Bond (Pi-Donor) | 3.1124 | | |
| Ser237:OG - EA | Other (Pi-lone pair) | 2.9816 | | |
| Tyr105 - EA | Hydrophobic (Pi-Pi stacked) | 5.7935 | | |
| Tyr105 - EA | Hydrophobic (Pi-Pi stacked) | 4.9257 | | |
| Tyr105 - EA | Hydrophobic (Pi-Pi stacked) | 4.1056 | | |
| **Cefotaxime (CTX)** | | | | |
| Lys234:HZ3 - CTX:O | Hydrogen Bond + Electrostatic | 2.7643 | −7.022 | $1.41 \times 10^5$ |
| Asn104: HD22 - CTX:O | Hydrogen Bond | 1.8275 | | |
| Asn104: HD22 - CTX:O | Hydrogen Bond | 2.6489 | | |
| Asn132: HD22 - CTX:O | Hydrogen Bond | 2.4406 | | |
| Asn170: HD21 - CTX:O | Hydrogen Bond | 2.8474 | | |
| Ser237:H - CTX:O | Hydrogen Bond | 2.1126 | | |
| Thr235:HG1 – CTX:O | Hydrogen Bond | 2.1716 | | |
| Thr235:HG1 – CTX:O | Hydrogen Bond | 2.1235 | | |
| Ser70:HG – CTX:O | Hydrogen Bond | 2.4821 | | |
| Ser70:HB2 - CTX:O | Carbon Hydrogen Bond | 2.4650 | | |
| Ser70:HB2 - CTX:O | Carbon Hydrogen Bond | 2.6008 | | |
| Thr216:HB - CTX:O | Carbon Hydrogen Bond | 2.8962 | | |
| CTX:H - PRO167:O | Carbon Hydrogen Bond | 2.2561 | | |
| Ser237:O - CTX | Electrostatic (Pi-Anion) | 3.2239 | | |
| Tyr105 - CTX | Hydrophobic (Pi-Alkyl) | 4.3816 | | |

152. The analysis of physiochemical properties of the shortlisted compounds has shown that amongst 5 compounds, only 3 (Inositol, Guanosine and Ellagic acid) obeyed Lipinski's rule [40]. The MM-GBSA calculations has shown that Ellagic acid had the lowest MM-GBSA energy (−49.55 kcal $mol^{-1}$) followed by Naringin dihydrochalcone (−46.38 kcal $mol^{-1}$), Salvianolic acid B (−45.42 kcal $mol^{-1}$), Guanosine (−35.14 kcal $mol^{-1}$) and Inositol (−33.34 kcal $mol^{-1}$) for binding to CTX-M-152. The lowest MM-GBSA energy of Ellagic acid suggest a strong interaction of Ellagic acid with CTX-M-152 modelled protein. The MM-GBSA of Naringin dihydrochalcone and Salvianolic acid were also close to that of Ellagic acid, but these were not considered for further evaluation owing to their deviation from Lipinski's rule of five.

Ellagic acid is a versatile phenolic compound found in pomegranate, strawberry, raspberry and grapes. It can also be extracted from the stem of *Eucalyptus globulus* and bark of *Eucalyptus maculate* [41]. Pomegranate extracts are known for its versatility since the antiquity. It is well known for its remedial effects against diarrhoea, infections caused by microbes, acidosis, and dysentery [42]. Pomegranate extracts show synergic effects in combination with Ciprofloxacin against ESBL producers such as *Escherichia coli, Klebsiella pneumoniae*, and metallo β-lactamase producer *Pseudomonas aeruginosa* [43]. Liposomal formulation of Cefotaxime with Ellagic acid has been proved to be a good antimicrobial agent against *Staphylococcus aureus* and *Bacillus subtilis* (both are Gram positive), *Escherichia coli* and *Pseudomonas aeruginosa* (Gram negative), with enhanced permeability of Cefotaxime [44]. Ellagic acid is also known to exert inhibitory action against biofilm formation in different bacteria [45]. It has also been reported to exhibit inhibitory activity against *Mycobacterium tuberculosis* and ESBL/KPC type carbapenemase producers such as *Klebsella pneumoniae* [46]. The mechanistic insight into the antimicrobial activity of Ellagic acid has shown that it inhibits DNA primase (DnaG) and thus inhibits the bacterial replication [47]. The results of the study revealed Ellagic acid as equally competent in inhibiting the growth of *Kluyvera georgiana* isolates harbouring *bla*CTX-M gene. The zone of inhibition of CTX-M-152 towards Cefotaxime shows similarity with Ellagic acid; thereby suggest Ellagic acid as equally compatable in terms of inhibiting growth of *Kluyvera georgiana*. It is worth to notice that both Ellagic acid and Cefotaxime had occupied amino acid residues critical for the hydrolytic activity of CTX-M-152. The residues such as Ser70, Asn104, Tyr105, Asn132, Asn170, Thr216, Thr235 and Ser237 were commonly involved in the interaction with Ellagic acid as well as Cefotaxime. It is worth to note that these residues have been reported to play crucial role in maintaining the active site of CTX-M-152 and hydrolysis of β-lactam antibiotics. In this study, we have shown a different mechanism wherein Ellagic acid binds to β-lactamase and thus can check the growth of bacteria. It is inferred that if CTX-M-152 is capable of efficiently hydrolyzing Cefotaxime as reported for other CTX-M type enzymes such as CTX-M-15, CTX-M-16, CTX-M-25, and CTX-M-18 [48,49], then Ellagic acid holds strong promise in combatting resistance among bacterial isolates that harbor CTX-M-152 or its related enzymatic moieties.

## 5. Conclusion

Here, we report the identification of two *Kluyvera georgiana* isolates harboring $bla_{\text{CTX-M-152}}$ gene. A three-dimensional model of CTX-M-152 protein is generated using CTX-M-14 (PDB Id: 4UA6) as template, and characterized by various tools such as SAVES, Ramachandran plot, ProsaII, Procheck, Verify 3D, RMSDs, Molprobity clash score. In a quest to identify novel inhibitors against *bla*CTX-M-152 harboring bacteria, we have employed a computational approach to screen a library of natural compounds, containing 803 unique and structurally diverse compounds. We identified 5 compounds (Naringin dihydrochalcone, Salvianolic acid B, Inositol, Guanosine and Ellagic acid) with a XP docking score of ≤ −6 kcal $mol^{-1}$ are identified as a potential inhibitor of CTX-M-152. The analysis of these shorthlisted compounds by the change in free energy (MM-GBSA) and Lipinski's rule of five, leads to the

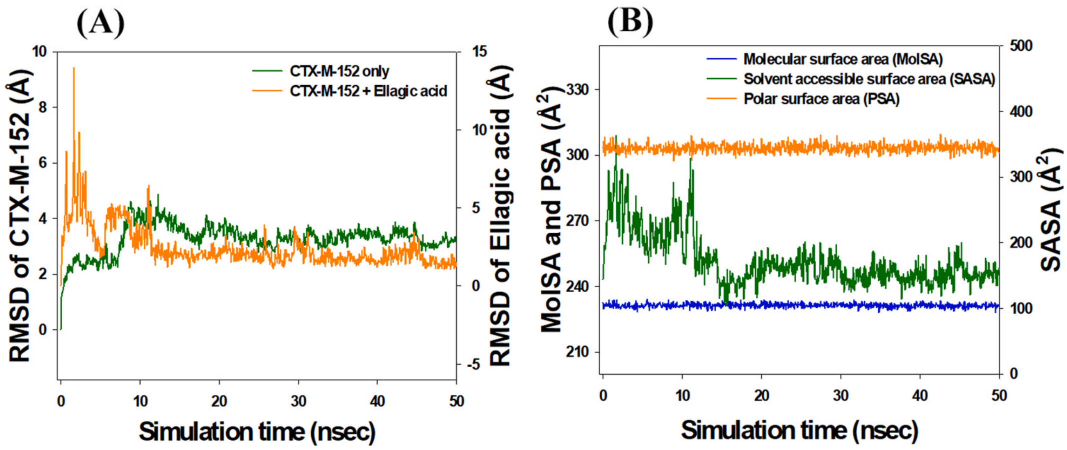



**Fig. 5.** (A) Molecular dynamic simulation of CTX-M-152 alone and in the presence of Ellagic acid, and (B) Changes in surface area of CTX-M-152 in the presence of Ellagic acid as a course of simulation time.

identification of Ellagic acid as the most potential inhibitor of CTX-M-152. Further, the stability of CTX-M-152-Ellagic acid complex and the nature of interaction between them is confirmed by molecular dynamics simulation. In this study, we have shown a different mechanism wherein Ellagic acid binds to β-lactamase and thus can check the growth of bacteria. It is inferred that if CTX-M-152 is capable of efficiently hydrolyzing Cefotaxime as reported for other CTX-M type enzymes such as CTX-M-15, CTX-M-16, CTX-M-25, and CTX-M-18 [50,51], then Ellagic acid holds strong promise in combatting resistance among bacterial isolates that harbor CTX-M-152 or its related enzymatic moieties. This is the first study reporting Ellagic acid as a novel inhibitor of CTX-M-152. It could serve as a starting points for the synthesis of more potent inhibitors against ESBLs. The findings of this study can be used to understand the mechanisms of resistance mediated by CTX-M-152, which in turn will help in formulating new strategies for preventing their transmission as part to combat the growing menace of antibiotic resistance.

## Author credit statement

Conceptualization, A.T.J., S.R. and A.T.; methodology, H.L.; software, H.L., M.A.B. and V.K; validation, M.Z.A., A.S.A. and M.S.A.; formal analysis, S.R. and A.T.; writing-original draft preparation, H.L., M.A.B. and V.K.; writing—editing, A.T.J., S.R. and A.T.; supervision, A. T.; funding acquisition, M.Z.A., A.S.A. and M.S.A. All authors have read and agreed to the published version of the manuscript.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.micpath.2020.104688.

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